

1. What is the purpose of a decision tree in machine learning?

- A. To reduce data storage
- B. To improve graphics performance
- C. To make decisions by learning rules from data
- D. To compress datasets

Answer: C

2. In a decision tree, what does a leaf node represent?

- A. A question
- B. A feature
- C. The result/output
- D. A split

Answer: C

3. What is the measure used by the ID3 algorithm to choose the best attribute?

- A. Gini Index
- B. Variance
- C. Information Gain
- D. Correlation

Answer: C

4. Which algorithm is known for using the Gini Index for splitting?

- A. ID3
- B. C4.5
- C. CART
- D. SVM

Answer: C

5. What is the main goal of pruning in a decision tree?

- A. To increase the number of branches
- B. To reduce tree depth
- C. To improve accuracy on training data
- D. To prevent overfitting

Answer: D

6. Name two types of pruning techniques.

Answer: Pre-pruning and Post-pruning

7. Which of the following is a disadvantage of decision trees?

- A. Difficult to interpret
- B. Need for feature scaling
- C. High risk of overfitting
- D. Cannot handle categorical data

Answer: C

8. What is the key improvement of C4.5 over ID3?

- A. Uses Gini Index
- B. Supports pruning and continuous attributes
- C. Builds binary trees only
- D. Doesn't require data

Answer: B

9. What is a Random Forest?

- A. A single decision tree
- B. A rule-based system
- C. A group of decision trees combined to improve accuracy
- D. A clustering algorithm

Answer: C

10. True or False: Decision Trees require data normalization before training.

Answer: False

Hierarchical Clustering - MCQ Practice Set

1. Which of the following is not a type of hierarchical clustering?

- A. Agglomerative
- B. Divisive
- C. K-Means
- D. None of the above

Answer: C. K-Means

2. In agglomerative clustering, the process starts with:

- A. One big cluster
- B. Predefined clusters
- C. Each point as a separate cluster
- D. Randomly chosen centroids

Answer: C. Each point as a separate cluster

3. Which of the following methods uses the minimum distance between points in two clusters?

- A. Complete Linkage
- B. Average Linkage
- C. Single Linkage
- D. Ward's Method

Answer: C. Single Linkage

4. What is the purpose of a dendrogram?

- A. Visualize data distribution
- B. Show hierarchical clustering process
- C. Identify centroids
- D. Measure accuracy

Answer: B. Show hierarchical clustering process

5. Which linkage method tends to produce compact, spherical clusters?

- A. Ward's Method
- B. Single Linkage
- C. Complete Linkage
- D. Divisive Method

Answer: A. Ward's Method

6. When should you cut the dendrogram?

- A. When the clusters are equally sized
- B. Where the vertical distance is smallest
- C. At the point of maximum distance between merges
- D. Just above the root

Answer: C. At the point of maximum distance between merges

7. Hierarchical clustering is best suited for:

- A. Large datasets with millions of rows
- B. Real-time streaming data
- C. Small to medium datasets with clear structure
- D. Training neural networks

Answer: C. Small to medium datasets with clear structure

8. Which clustering algorithm does not require specifying the number of clusters beforehand?

- A. K-Means
- B. Hierarchical Clustering
- C. DBSCAN
- D. Both B and C

Answer: D. Both B and C

1. **What does the 'K' in KNN stand for?**

Answer: C. Number of Nearest Neighbors

2. **Which type of learning algorithm is KNN?** **Answer:** A. Supervised

3. **KNN is mostly used for which of the following?**

Answer: C. Both A and B

(KNN can be used for classification and regression tasks, though it's more common for classification.)

4. **What happens when you increase the value of K in KNN?** **Answer:** B. Model becomes simpler and may underfit

5. **Which distance metric is commonly used in KNN?**

Answer: D. All of the above

(Euclidean is most common, but others like Manhattan and Cosine Similarity can also be used.)

6. **KNN requires a training phase like other algorithms. True or False?**

Answer: False

(KNN is a lazy learner — it doesn't train the model but uses the entire dataset at prediction time.)

7. **Why is feature scaling important in KNN?**

Answer: B. It avoids dominance of one feature due to scale

8. **Which of the following is a drawback of KNN?**

Answer: D. Both B and C

(KNN has high prediction time and requires storing the full dataset.)

9. **What happens if there is a tie in the nearest neighbors' class labels?**

Answer: D. Depends on the implementation

(Some implementations break ties randomly, others by smallest label, etc.)

10. **What does KNN use to determine the class of a data point?**

A. Neural networks

B. Decision trees

C. The majority class of its K nearest neighbors

D. Statistical probability **Answer:** C

11. **Which of the following is NOT a typical use case of KNN?**

A. Handwriting recognition

B. Stock price prediction

C. Spam detection

D. Clustering

Answer: D

12. **In KNN, what does a smaller value of K typically lead to?**

A. More stable predictions

B. More noise sensitivity

C. Better generalization

D. Increased underfitting **Answer:** B

13. If you set $K = 1$ in KNN, the model will always:

- A. Underfit the data
- B. Overfit the data
- C. Ignore distance
- D. Perform feature selection **Answer: B**

14. What is the main disadvantage of using a very high value for K ?

- A. Increased overfitting
- B. High computational cost
- C. Underfitting and reduced sensitivity to local patterns
- D. The model won't work **Answer: C**

15. How does KNN handle multi-class classification?

- A. One-vs-All
- B. Logistic Regression
- C. Picks the majority class from K nearest points
- D. It can't handle multi-class classification

Answer: C

16. What is a weighted KNN?

- A. KNN where each feature is weighted equally
- B. KNN that adjusts learning rate
- C. KNN where closer neighbors contribute more to the vote than farther ones
- D. A deep learning model **Answer: C**

17. Which of the following is true about KNN's time complexity during prediction?

- A. It is constant time $O(1)$
- B. It is logarithmic $O(\log n)$
- C. It is linear $O(n)$
- D. It is quadratic $O(n^2)$ **Answer: C**

18. KNN performs poorly when:

- A. Data has very low dimensions
- B. The dataset is small
- C. Data is not labeled
- D. Features are not normalized and have varying scales **Answer: D**

19. Which of the following strategies can be used to select the optimal value of K ?

- A. Grid Search with Cross-Validation
- B. Setting $K = 1$ always
- C. Using unsupervised learning
- D. Random guessing **Answer: A**

KNN Algorithm Quiz with Answers

Basic Level

1. What does the 'K' in KNN stand for?

Answer: C. Number of Nearest Neighbors

2. Which type of learning algorithm is KNN?

Answer: A. Supervised

3. KNN is mostly used for which of the following?

Answer: C. Both A and B

Intermediate Level

4. What happens when you increase the value of K in KNN?

Answer: B. Model becomes simpler and may underfit

5. Which distance metric is commonly used in KNN?

Answer: D. All of the above

6. KNN requires a training phase like other algorithms. True or False?

Answer: False

Advanced Level

7. Why is feature scaling important in KNN?

Answer: B. It avoids dominance of one feature due to scale

8. Which of the following is a drawback of KNN?

Answer: D. Both B and C

9. What happens if there is a tie in the nearest neighbors' class labels?

Answer: D. Depends on the implementation

1. In a decision tree, what does a leaf node represent?
(a) A split point (b) A decision rule
(c) A final outcome or prediction (d) None of the above
-> (c) A final outcome or prediction
2. What is the purpose of dimensionality reduction in unsupervised learning?
(a) To increase computational complexity (b) To reduce the number of features
(c) To add noise to the data (d) None of the above
-> (b) To reduce the number of features
3. Which of the following is NOT a dimensionality reduction technique?
(a) Principal Component Analysis (PCA) (b) Linear Regression
(c) t-Distributed Stochastic Neighbor Embedding (tSNE)
(d) Singular Value Decomposition (SVD)
-> (b) Linear Regression
4. Which of the following is NOT a common application of unsupervised learning? (a) Image classification (b) Anomaly detection
(c) Market segmentation (d) Customer recommendation systems
-> (a) Image classification
5. What is the primary difference between supervised and unsupervised learning? (a) Supervised learning requires labeled data; unsupervised learning does not.
(b) Unsupervised learning requires labeled data; supervised learning does not.
(c) Both supervised and unsupervised learning require labeled data.
(d) Both supervised and unsupervised learning do not require labeled data.
-> (a) Supervised learning requires labeled data; unsupervised learning does not.
6. Which of the following library is used for boosting generalized additive models?
(a) GamBoost (b) Gbm
(c) Ada (d) All of these
-> (a) GamBoost
7. Let's say, you are using activation function X in hidden layers of neural network. At a particular neuron for any given input, you get the output as "-0.0001". Which of the following activation function could X represent?
(a) ReLU (b) Tanh
(c) Sigmoid (d) All of these
-> (b) Tanh
8. What does the confusion matrix represent in classification tasks?
(a) The performance of the model in terms of true positives, true negatives, false positives, and false negatives
(b) The correlation between features
(c) The distribution of data points
(d) None of the above
-> (a) The performance of the model in terms of true positives, true negatives, false positives, and false negatives
9. What is the primary goal of reinforcement learning?
(a) To classify input data into predefined categories
(b) To maximize cumulative rewards by learning optimal policies
(c) To generate new examples similar to the training data

- (d) To cluster data points into compact groups
-> (b) To maximize cumulative rewards by learning optimal policies

10. Which of the following is NOT a component of a reinforcement learning system?

- (a) Reward function (b) Value function
(c) Policy (d) Training dataset
-> (d) Training dataset

5 MARKS QUESTIONS

1. Define Machine Learning and explain its significance in the field of computer science.

Machine Learning is a subset of **Artificial Intelligence** that allows computers to learn from data without being explicitly programmed.

Significance of Machine Learning in Computer Science:

- **Automation of Decision-Making:** ML allows systems to make intelligent decisions autonomously.
- **Predictive Analysis:** ML is widely used in making predictions, such as stock market forecasting, healthcare diagnostics, and sales forecasting.
- **Natural Language Processing (NLP):** ML powers applications like chatbots, speech recognition, and translation services by helping systems understand and generate human language.
- **Image and Video Recognition:** ML has revolutionized computer vision tasks, such as facial recognition, object detection, and autonomous vehicles.
- **Personalization:** ML is crucial in recommendation systems, where platforms like Netflix, YouTube, and Amazon use ML to recommend products or content tailored to the user's preferences.
- **Healthcare and Biomedicine:** ML helps in analyzing medical data for early disease detection, personalized treatment plans, and drug discovery.
- **Data Analysis and Insights:** ML algorithms help extract meaningful patterns and insights from vast amounts of data, which are crucial in industries like finance, marketing, and logistics.
- **Optimization:** From supply chain management to route planning, ML helps optimize processes and reduce costs in various domains.

2. Discuss the Polynomial Regression with a proper example.

Polynomial Regression is a type of regression analysis that models the relationship between the independent variable (X) and the dependent variable (Y) as an **nth-degree polynomial**. It is an extension of linear regression where we fit a polynomial equation to the data instead of a straight line.

In polynomial regression, the model tries to find a relationship between the variables by fitting a curve rather than a straight line.

General Polynomial Regression Equation:

The general equation for a polynomial regression model of degree n is:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \dots + \beta_n x^n + \epsilon \quad \text{Where:}$$

- y is the dependent variable (output).

- x is the independent variable (input).
- $\beta_0, \beta_1, \dots, \beta_n$ are the coefficients (weights) to be learned by the model.
- ϵ is the error term or residuals.
- n is the degree of the polynomial (which determines the complexity of the curve).

Example:

Imagine you're trying to predict the price of a car based on its age. A linear regression model might suggest a constant decrease in price as the car ages, which isn't realistic. A polynomial regression model, particularly one with a quadratic term (x^2), can capture the initial rapid depreciation followed by a slower decline later on.

3. Explain the concept of overfitting in Machine Learning.

Overfitting in machine learning occurs when a model learns the training data too well, including irrelevant details and noise, leading to poor performance on unseen data. Essentially, the model "memorizes" the training set rather than generalizing the underlying patterns. This results in high accuracy on the training data but low accuracy on new data.

Why Does Overfitting Happen?

Overfitting happens when the model becomes too complex, often due to:

- **Too many features:** A model with a large number of input features may learn intricate patterns that are not present in general data.
- **Excessive model complexity:** Highly complex models, like very deep neural networks or high-degree polynomials, may fit the training data perfectly, even fitting random fluctuations or noise.
- **Insufficient training data:** When the amount of training data is small, the model can easily memorize it without learning the underlying patterns that generalize.
- **Low regularization:** Regularization techniques like L1/L2 regularization are used to constrain the model and prevent overfitting. If regularization is not applied or is weak, the model can become overly complex.

Signs of Overfitting

- **High performance on training data:** The model shows very high accuracy, low error, or perfect predictions on the training set.
- **Low performance on validation/test data:** The model's performance drops significantly on unseen data or the test set, indicating poor generalization.

4. Machine Learning can perform tasks that are too complex for a person to directly handle" - justify the statement with proper example.

The statement "Machine Learning can perform tasks that are too complex for a person to directly handle" highlights one of the most significant advantages of machine learning (ML) in the real world. ML allows us to automate and optimize tasks that are difficult, time-consuming, or impossible for humans to perform manually, especially when dealing with massive amounts of data, complex patterns, or tasks requiring real-time decisions.

Here's a detailed justification of the statement with **examples**:

- **Data Volume:**
Humans can process a limited amount of information at a time. Machine learning algorithms can analyze massive datasets, identifying subtle correlations and patterns that would be missed by human analysis.
- **Pattern Recognition:**
Machine learning algorithms can identify patterns in data that humans might find difficult or impossible to see, allowing them to make predictions and decisions based on these patterns.
- **Learning from Examples:**
Machine learning algorithms can learn from examples and improve their performance over time as they are exposed to more data.
- **Automation:**
Machine learning can automate tasks that would be tedious, time-consuming, and errorprone for humans, freeing up human resources for more complex tasks.

5. Define precision, recall, and F1-score. Justify how those metrics used to evaluate classification models?

Precision measures the **accuracy of the positive predictions**.

Mathematically:

$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$

- **True Positives (TP):** The number of instances correctly classified as positive.
- **False Positives (FP):** The number of instances incorrectly classified as positive.

Recall measures the **ability of the model to identify all the actual positives**.

Mathematically:

$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$

- **True Positives (TP):** The number of instances correctly classified as positive.
- **False Negatives (FN):** The number of instances incorrectly classified as negative. The **F1-Score** is the **harmonic mean** of precision and recall.

Mathematically:

$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

Evaluating Classification Models Using Precision, Recall, and F1-Score:

Precision:

- Important when **false positives** are costly.
- Example: In email spam detection, you want to avoid classifying important emails as spam (false positives).
- **Use Case:** Medical tests where false positives could lead to unnecessary treatments.

Recall:

- Important when **false negatives** are costly.
- Example: In disease detection, you want to identify as many patients with a disease as possible, even if some of the diagnosed cases are false positives.
- **Use Case:** Fraud detection, where failing to catch fraud (false negative) can lead to financial losses.

F1-Score:

- Useful when both **precision and recall are important**.
- **Use Case:** A balance of precision and recall is needed, especially in cases like spam detection or medical diagnostics where both false positives and false negatives matter.

6. Define Neural Network. What's the trade-off between bias and variance?

A **Neural Network** is a computational model inspired by the way biological neural networks process information. Neural networks are used in machine learning to model complex patterns and relationships in data.

Bias-Variance Trade-off:

Bias is the difference between the model's prediction and the true values.

Variance refers to the model's sensitivity to small fluctuations in the training data.

- **High Bias and Low Variance (Underfitting):** If the model is too simple, it will have high bias and low variance.
- **Low Bias and High Variance (Overfitting):** If the model is too complex, it will have low bias but high variance. The model fits the training data very well, including noise and outliers, but performs poorly on new data.
- **Optimal Balance:** The goal is to find a model that achieves a balance between bias and variance, minimizing both. This typically results in a model that generalizes well to unseen data, achieving good predictive performance on both training and test datasets.

7. Write down 5 Major differences between regression and classification.

Here are 5 major differences between **regression** and **classification**:

Aspect	Regression	Classification
Output	Continuous numerical value	Discrete class label
Problem Type	Predicting a quantity or amount	Categorizing data into predefined classes
Evaluation Metrics	MSE, RMSE, MAE, R^2	Accuracy, Precision, Recall, F1-Score, Confusion Matrix
Output Range	Can be any real number	Limited to a set of classes
Algorithms	Linear Regression, Decision Trees (regression), Random Forest (regression), etc.	Logistic Regression, SVM, Decision Trees (classification), Random Forest (classification), etc.

8. Classification is better than regression" - Justify the statement with a proper example.

The statement "Classification is better than Regression" is **not universally true**.

Whether classification is better than regression depends entirely on the **nature of the**

problem being solved, the type of data available, and the objective of the model. **Both classification and regression** are important and used in different scenarios.

Key Reasons Why Classification Could Be Better Than Regression in These Cases:

- **Discrete Output:** If the target variable consists of discrete categories or labels, classification is naturally more suitable because it is designed to predict categories, whereas regression is meant for continuous outputs.
- **Interpretability:** Classification provides outputs that are easier to interpret in the context of decision-making. Regression might output a continuous score, but the interpretation is not as straightforward when the task requires distinct categorical outcomes.
- **Model Output:** In some cases, classification algorithms can also provide probability scores, which can be very useful for decision-making. Regression would not directly provide this information.

9. What is the utility of cross validation in machine learning?

Cross-validation in machine learning is a crucial technique used to assess how well a model will generalize to unseen data and prevent overfitting. It involves splitting the data into multiple subsets (folds), training the model on some subsets, and then evaluating its performance on the remaining subset. This process is repeated with different combinations of training and validation sets, providing a more robust estimate of the model's performance than a single train/test split.

Key utilities of cross-validation in machine learning are:

- **Improved Model Evaluation:**
Cross-validation helps in assessing the model's performance by training it on different subsets of the dataset. It splits the data into multiple training and testing sets, which ensures that the model is evaluated on different portions of the data.
- **Reduces Overfitting:**
Overfitting occurs when a model performs well on training data but poorly on unseen data. By using cross-validation, you get multiple validation sets, which helps identify models that might be overfitting.
- **Better Use of Data:**
Cross-validation allows for a better utilization of the available data. In traditional training-test splits, a large portion of the data is used only for training, while a smaller portion is used for testing. Cross-validation helps ensure that each data point is used for both training and testing.
- **More Reliable Performance Metrics:**
Cross-validation provides a more robust estimate of the model's performance. By training and testing the model on different data subsets, you get multiple performance scores.

10. In a dataset with 500 samples, 200 of which belong to class A and 300 belong to class B, what are the conditional probabilities of $P(A/B)$ and $P(B/A)$?

$P(A/B)$: Probability that a sample is in **class A** given that it's known to be in **class B**. $P(B/A)$: Probability that a sample is in **class B** given that it's known to be in **class A**. Given:

- Total samples = 500
- Samples in class A = 200 $\rightarrow P(A)=200 / 500=0.4$
- Samples in class B = 300 $\rightarrow P(B)=300 / 500=0.6$

But **without additional context or overlapping class membership**, computing:

- $P(A|B) = P(A \cap B) / P(B)$ • $P(B|A) = P(A \cap B) / P(A)$ is **not defined**, because $P(A \cap B) = 0$ if the classes are disjoint.

If A and B are **mutually exclusive**, then:

- $P(A|B) = 0$,
- $P(B|A) = 0$

11. What are the differences between Soft Clustering and Hard Clustering? Give Example.

Feature	Hard Clustering	Soft Clustering
Membership Assignment	Each data point belongs to only one cluster	Each data point can belong to multiple clusters with varying degrees
Example Algorithm	K-Means, DBSCAN	Gaussian Mixture Model (GMM), Fuzzy C-Means
Cluster Assignment	Binary (either in a cluster or not)	Probabilistic (assigned a probability to each cluster)
Cluster Overlap	No overlap between clusters	Overlap is allowed between clusters
Handling Ambiguity	Does not handle ambiguity well	Handles ambiguity and fuzzy boundaries well
Use Cases	Clearly separated clusters	Overlapping clusters or uncertain boundaries
Computational Complexity	Less complex	More complex due to probability calculations

Examples:

Hard Clustering:

K-Means Clustering: This algorithm assigns each data point to one of the k clusters based on its distance to the nearest centroid.

Soft Clustering:

Fuzzy C-Means (FCM): A clustering algorithm where each point has a degree of membership to all clusters, with fuzzy partitions.

12. Explain the coefficient of determination (R^2) in linear regression?

The **coefficient of determination (R^2)** is a key metric in linear regression that measures the proportion of the variance in the dependent variable that is predictable from the independent variables. It provides an indication of how well the regression model fits the data.

The **coefficient of determination (R^2)** is defined as:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Where:

- y_i = Actual value of the dependent variable for the i-th data point.
- $\hat{y}_i$ = Predicted value of the dependent variable for the i-th data point based on the model.
- $\bar{y}$ = Mean of the actual values of the dependent variable across all data points.

Interpretation of R^2 :

- **$R^2=1$:** The model explains **100%** of the variance in the dependent variable. This means that all data points lie exactly on the regression line, and the model has perfect predictive power.
- **$R^2=0$:** The model explains **0%** of the variance in the dependent variable. This means the model performs no better than simply predicting the mean of the dependent variable for all data points.
- **$R^2<0$:** In some cases, the R^2 value can be negative, which typically indicates that the model is worse than a simple mean-based model. This happens when the model fits the data poorly, and its predictions are worse than just predicting the average value of the target variable.

ML Classification Error Metrics Quiz

1. What does a confusion matrix display?

- A) Only true positives
- B) The relationship between predicted and actual labels**
- C) Regression errors
- D) Only accuracy

2. Which metric measures how many actual positive cases were correctly identified by the model?

- A) Precision
- B) Accuracy
- C) Specificity
- D) Sensitivity**

3. If a model has high precision but low recall, what does that mean?

- A) It predicts too many negatives
- B) It rarely misses actual positives
- C) It makes many false positives
- D) It correctly identifies positives but misses many**

4. What does the F1-score represent?

- A) Average of Sensitivity and Specificity
- B) Harmonic mean of Precision and Recall**
- C) Product of Precision and Recall
- D) Sum of TP and TN

5. Which metric would you use if false negatives are more harmful than false positives?

- A) Specificity
- B) Recall**

C) Accuracy

D) FPR

6. What is the best AUC value a model can achieve?

A) 0.5

B) 1

C) 0

D) Depends on FPR

7. Which curve is better to evaluate models with imbalanced datasets?

A) ROC Curve

B) Precision-Recall Curve

C) Confusion Matrix

D) AUC Curve

8. In the ROC curve, what does the Y-axis represent?

A) Precision

B) False Positive Rate

C) True Positive Rate

D) Accuracy

Neural Networks Quiz with Answers

1. Which of the following best describes a perceptron?

- A basic linear classifier with learnable weights

2. Why can't a single perceptron solve the XOR problem?

- XOR is not linearly separable

3. What is the main purpose of an activation function?

- To introduce non-linearity into the model

4. Which activation function outputs values between 0 and 1?

- Sigmoid

5. In a neural network, what does the bias term do?

- Allows the activation to shift left or right

6. What happens during forward propagation?

- Output is predicted by passing inputs through the network

7. What is the goal of backpropagation?

- To compute gradients for each parameter

8. In the loss function for classification, what is used after the final layer?

- Softmax

9. What is cross-entropy used for in neural networks?

- Comparing predicted and true probability distributions

10. What does gradient descent do?

- Updates weights to minimize loss

11. What is the output of softmax always?

- Values between 0 and 1 that sum to 1

12. What type of function is used for the output layer in regression tasks?

- Linear

13. Which of the following is a valid loss function for regression?

- Mean Squared Error

14. What does entropy measure?

- Degree of randomness or uncertainty

15. When does training stop in the perceptron learning algorithm?

- When all inputs are classified correctly

Supervised Learning - MCQ Quiz

1. Which of the following is NOT a type of supervised learning?

- a) Classification
- b) Clustering
- c) Regression
- d) Binary classification

Answer: b) Clustering

2. What does the slope 'm' represent in linear regression?

- a) Output label
- b) Cost function
- c) Rate of change
- d) Intercept

Answer: c) Rate of change

3. Which function is used in logistic regression?

- a) Tanh
- b) ReLU
- c) Sigmoid
- d) Linear

Answer: c) Sigmoid

4. Entropy in decision trees measures:

- a) Distance between nodes
- b) Number of leaves

Supervised Learning - MCQ Quiz

- c) Uncertainty or disorder
- d) Tree depth

Answer: c) Uncertainty or disorder

5. Which metric is most useful for imbalanced classification problems?

- a) Accuracy
- b) Precision
- c) F1 Score
- d) Mean Squared Error

Answer: c) F1 Score

6. Gradient Descent helps to:

- a) Increase error
- b) Make the model more complex
- c) Minimize the cost function
- d) Visualize the data

Answer: c) Minimize the cost function

7. What does a high variance model indicate?

- a) Underfitting
- b) Overfitting
- c) Perfect generalization
- d) Slow training

Answer: b) Overfitting

Supervised Learning - MCQ Quiz

8. Cross-validation helps in:

- a) Data scaling
- b) Model visualization
- c) More reliable model evaluation
- d) Feature elimination

Answer: c) More reliable model evaluation

9. Which technique scales data to have zero mean and unit variance?

- a) Normalization
- b) Random scaling
- c) Standardization
- d) Data shuffling

Answer: c) Standardization

10. Bias-Variance Tradeoff aims to:

- a) Increase both bias and variance
- b) Eliminate training error only
- c) Balance model simplicity and accuracy
- d) Maximize entropy

Answer: c) Balance model simplicity and accuracy

Basic Level

1. **What is the main goal of a Support Vector Machine (SVM)?**

- A) Minimize the number of features
- B) Maximize the number of support vectors
- C) Maximize the margin between classes
- D) Minimize the number of classes

Answer: C) Maximize the margin between classes

2. **Which of the following are called “support vectors” in SVM?**

- A) All data points
- B) Data points farthest from the hyperplane
- C) Data points that define the margin
- D) Randomly selected data points

Answer: C) Data points that define the margin

3. **In SVM, the area between the decision boundary and the nearest data points is called:**

- A) Cost area
- B) Margin
- C) Threshold
- D) Kernel

Answer: B) Margin

Intermediate Level

4. **What kind of loss function does SVM use?**

- A) Squared loss
- B) Log loss
- C) Hinge loss
- D) Cross-entropy loss

Answer: C) Hinge loss

5. **Which of the following optimization techniques is used in training SVM?**

- A) Newton-Raphson
- B) Stochastic Gradient Descent
- C) Lagrange Multipliers
- D) Genetic Algorithm

Answer: C) Lagrange Multipliers

6. The final decision function in a linear SVM is of the form:

- A) $y = w^2 + b$
- B) $y = w^T x + b$
- C) $y = w + x$
- D) $y = x^T b + w$

Answer: B) $y = w^T x + b$

Advanced Level

7. What do the Karush-Kuhn-Tucker (KKT) conditions help identify in SVM?

- A) Features with high correlation
- B) Classes with low accuracy
- C) Optimal solutions for constraints
- D) Maximum number of support vectors

Answer: C) Optimal solutions for constraints

8. In SVM's objective function, the regularization term $\lambda \|w\|_2^2$ helps:

- A) Increase overfitting
- B) Encourage complex models
- C) Minimize the margin
- D) Control model complexity

Answer: D) Control model complexity

1. A single McCulloch Pitts neuron can be used to represent boolean functions that are linearly separable. what about non-boolean inputs?

The McCulloch-Pitts model was originally designed for **binary (Boolean)** input and output, typically {0, 1}. However, when extended to **non-Boolean (continuous or real-valued)** inputs, the classic MP neuron **cannot handle them directly** without modification.

To handle **non-Boolean inputs**, we typically transition from the MP neuron to the **Perceptron** or **modern artificial neurons**, which:

- Accept real-valued inputs.
- Use learnable weights and biases.
- Often apply **differentiable activation functions** like sigmoid, tanh, or ReLU instead of a step function.

2. Do we always need to hand-pick the threshold?

In the original McCulloch-Pitts neuron:

Yes — the threshold is **manually chosen**.

- The neuron **fires (outputs 1)** if the weighted sum of the inputs $\geq$ threshold.
- This threshold is **not learned**; it's set by the designer based on the logic function (like AND, OR, etc.).

In learning models (e.g., Perceptron):

No — you **do not** need to hand-pick the threshold.

- The threshold is replaced by a **bias term**.
- Both **weights and bias are learned** using algorithms like the **Perceptron Learning Rule** or **Gradient Descent**.

3. What if some inputs have more importance than others?

In a neural model like a **Perceptron**, **input importance** is captured by the **weights**.

The output is based on the **weighted sum** of inputs:

$$y = \text{activation}(w_1x_1 + w_2x_2 + \dots + w_nx_n + b)$$

Each **input x_i** is multiplied by a **weight w_i** .

- Inputs with **larger weights** have **more influence** on the output.
- Inputs with **small or zero weights** have **less or no effect**.

During learning:

- The model **adjusts weights** through training.
- Over time, it **automatically learns** which inputs are more important, by giving them higher weights.

4. What about functions that are not linearly separable?

These are functions where **no straight line** can separate the classes correctly.

Example: **XOR Function**

A **single McCulloch-Pitts neuron** or **single-layer perceptron** cannot represent non-linearly separable functions like XOR.

To solve this, we need a **Multi-Layer Perceptron** — also known as a **Feedforward Neural Network** with at least **one hidden layer**.

- The **hidden layer(s)** allow the model to learn **nonlinear decision boundaries**.
- Nonlinear **activation functions** (e.g., sigmoid, ReLU) are key to this ability.

With backpropagation and gradient descent, the network can learn:

- How to **combine multiple neurons** in hidden layers
- To represent **complex and non-linearly separable functions**

5. Describe one technique to reduce overfitting.

One effective technique to reduce **overfitting** is **regularization**.

Regularization adds a **penalty term** to the loss function during training, discouraging the model from learning overly complex patterns that fit the noise in the training data.

- **L1 regularization (Lasso)** adds the absolute value of the weights to the loss function. It can shrink some weights to zero, effectively performing feature selection.
- **L2 regularization (Ridge)** adds the square of the weights to the loss function. It discourages large weights, helping to keep the model simpler and more general.

6. Explain the role of the decision boundary in logistic regression.

In **logistic regression**, the **decision boundary** is a hypersurface that separates the data points into different classes based on the model's predicted probabilities.

Role of the Decision Boundary:

- Logistic regression models the **probability** that a given input belongs to a particular class.
- It uses the **sigmoid function** to output probabilities between 0 and 1.
- A **threshold** is set (commonly 0.5):
 - If the predicted probability $\geq 0.5 \rightarrow$ classify as class 1
 - If the predicted probability $< 0.5 \rightarrow$ classify as class 0

Mathematically, for a model:

$$\hat{y} = \frac{1}{1 + e^{-(w^T x + b)}}$$

The decision boundary is where:

$$w^T x + b = 0$$

7. Explain the significance of the diagonal line (AUC = 0.5) in an ROC curve. What does it indicate about a classifier?

Significance of the Diagonal Line (AUC = 0.5):

- It is the line from (0,0) to (1,1) in ROC space.
- **True Positive Rate (TPR)** increases proportionally with **False Positive Rate (FPR)**.
- The classifier performs **no better than random guessing**.

Interpretation:

- **AUC = 0.5** $\rightarrow$ Classifier has **zero predictive power**.
- It's a baseline; any useful classifier should achieve **AUC > 0.5**.
- **Closer the ROC curve is to the top-left corner**, the better the classifier.

8. The regularization parameter (C) in SVM controls the trade-off between maximizing the margin and minimizing misclassification. Explain how increasing or decreasing C affects the decision boundary.

♦ **High C:**

- The model **penalizes misclassification more heavily**.
- It tries to classify all training examples correctly, even if that results in a **narrower margin**.
- Decision boundary becomes **more complex**, potentially leading to **overfitting**.

♦ **Low C:**

- The model **allows more misclassifications** in favor of a **wider margin**.
- It prioritizes generalization over perfect training accuracy.
- Decision boundary becomes **smoother**, possibly **underfitting** the data if C is too small.

9. A decision tree classifier is trained on a dataset containing 30 samples, equally divided between Class A and Class B. After a split, the data is divided into Left Node with 12 samples (8 from Class A, 4 from Class B) and Right Node with 18 samples (7 from Class A, 11 from Class B). Calculate the entropy of the parent node before the split and both child nodes after the split, and determine the information gain from the split.

Entropy of the Parent Node:

Total samples: 30

Class A: 15

Class B: 15

Proportion: $p_A=15/30=0.5$, $p_B=0.5$

Entropy_{parent} = $-(0.5 \log_2 0.5 + 0.5 \log_2 0.5) = -(0.5 \cdot -1 + 0.5 \cdot -1) = 1.0$

Entropy of Left Child Node:

$$p_A = \frac{8}{12} = \frac{2}{3}, \quad p_B = \frac{4}{12} = \frac{1}{3}$$

$$\text{Entropy}_{\text{left}} = -\left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{1}{3} \log_2 \frac{1}{3}\right) = -\left(\frac{2}{3} \cdot (-0.585) + \frac{1}{3} \cdot (-1.585)\right) = 0.918$$

Entropy of Right Child Node:

$$p_A = \frac{7}{18}, \quad p_B = \frac{11}{18}$$

$$\text{Entropy}_{\text{right}} = -\left(\frac{7}{18} \log_2 \frac{7}{18} + \frac{11}{18} \log_2 \frac{11}{18}\right) = -(0.389 \cdot (-1.364) + 0.611 \cdot (-0.707)) \approx 0.964$$

Weighted Average Entropy After the Split:

Entropy_{children} = $12/30 \cdot 0.918 + 18/30 \cdot 0.964 = 0.3672 + 0.5784 = 0.9456$

Information Gain

Information Gain = Entropy_{parent} - Entropy_{children} = $1.0 - 0.9456 = 0.0544$

Final Answer:

- **Entropy of Parent Node:** 1.0
- **Entropy of Left Node:** ≈ 0.918
- **Entropy of Right Node:** ≈ 0.964
- **Information Gain:** ≈ 0.0544

10. A company wants to predict whether a customer will attend a product launch event based on two features: Job Role and Preferred Communication Channel. For a given dataset, if a new customer is a "Manager" and prefers "Email" as their communication channel, use the Naïve Bayes classifier to predict whether they will attend the event.

Calculate Prior Probabilities:

Event	Count
Yes	4
No	6
Total	10

CID	Job Role	Preferred Communication Channel	Attended Event (Yes/No)
1	Engineer	Phone	No
2	Manager	Email	Yes
3	Developer	Email	No
4	Manager	Email	Yes
5	Engineer	Email	No
6	Manager	Phone	Yes
7	Developer	Phone	No
8	Manager	Email	Yes
9	Engineer	Phone	No
10	Developer	Email	No

$P(\text{Yes})=4/10=0.4$, $P(\text{No})=6/10=0.6$

For "Manager":

- Out of 4 "Yes" entries:
 - Manager = 3 $\rightarrow P(\text{Manager}|\text{Yes})=3/4=0.75$
- Out of 6 "No" entries:
 - Manager = 0 $\rightarrow P(\text{Manager}|\text{No})=0/6=0$

For "Email":

- Out of 4 "Yes" entries:
 - Email = 3 $\rightarrow P(\text{Email}|\text{Yes})=3/4=0.75$
- Out of 6 "No" entries:
 - Email = 4 $\rightarrow P(\text{Email}|\text{No})=4/6\approx 0.667$

For Yes:

$$P(\text{Yes}) \cdot P(\text{Manager}|\text{Yes}) \cdot P(\text{Email}|\text{Yes}) = 0.4 \cdot 0.75 \cdot 0.75 = 0.225$$

For No:

$$P(\text{No}) \cdot P(\text{Manager}|\text{No}) \cdot P(\text{Email}|\text{No}) = 0.6 \cdot 0 \cdot 0.667 = 0$$

Since $0.225 > 0$, the **Naïve Bayes classifier predicts that the customer will attend the event.**

11. Derive the update rule for Gradient Descent when minimizing a quadratic cost function. Explain how the gradient influences the parameter updates.

Consider a simple linear regression scenario with one feature. The cost function (Mean Squared Error) is:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Where:

- $h_{\theta}(x) = \theta x$ is the hypothesis
- m is the number of training examples
- θ is the parameter to be learned

To simplify, suppose:

$$J(\theta) = 1/2(\theta x - y)^2$$

Compute the Gradient:

Take the derivative (gradient) of $J(\theta)$ with respect to θ :

$$dJ/d\theta = d(1/2(\theta x - y)^2)/d\theta = (\theta x - y)x$$

This is the **gradient**—it tells us the slope of the cost function at θ .

Gradient Descent Update Rule:

Gradient Descent updates parameters in the direction **opposite to the gradient**:

$$\Theta := \Theta - \alpha \cdot dJ/d\theta$$

Substitute the gradient:

$$\Theta := \Theta - \alpha(\theta x - y)x$$

Where α is the **learning rate**.

Role of the Gradient:

- The **gradient** indicates how much the cost would change if θ changes.
- If the gradient is **large**, the update is large $\rightarrow$ quick movement in parameter space.
- If the gradient is **small**, the update is small $\rightarrow$ slow, fine-tuned steps.
- The gradient being **positive or negative** tells whether to **decrease or increase** θ .

12. A medical research team is testing an ML model for diagnosing a disease. The model was tested on 200 patients, where 80 were actually sick. The model predicted 70 patients as sick, out of which 60 were truly sick, while 10 were healthy but misclassified. Additionally, 20 sick patients were misclassified as healthy. Calculate the accuracy, precision, recall, and F1-score of the model.

From the problem:

Total patients = 200

Actual sick (positive class) = 80

Model predicted sick = 70

- Of those, **60 were truly sick $\rightarrow$ True Positives (TP) = 60**
- **10 were healthy $\rightarrow$ False Positives (FP) = 10**

20 sick were misclassified as healthy $\rightarrow$ False Negatives (FN) = 20

True Negatives (TN) = Total - TP - FP - FN = 200 - 60 - 10 - 20 = 110

Accuracy = $(TP+TN) / (FP+FN+TP+TN) = (60+110) / 200 = 170 / 200 = 0.85$

Precision = $TP / (TP+FP) = 60 / (60+10) = 60 / 70 \approx 0.857$

Recall = $TP / (TP+FN) = 60 / (60+20) = 60 / 80 = 0.75$

F1_Score = $(2 \cdot \text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall}) = (2 \cdot 0.857 \cdot 0.75) / (0.857+0.75) = 2 \cdot 0.3998 = 0.7996 \approx 0.800$

13. State the difference between supervised and unsupervised learning.

Feature	Supervised Learning	Unsupervised Learning
Data Type	Labeled data	Unlabeled data
Goal	Predict output/target variable	Discover hidden patterns or structure in data

Feature	Supervised Learning	Unsupervised Learning
Output	Known (provided in training data)	Unknown (inferred from data)
Common Tasks	Classification, Regression	Clustering, Dimensionality Reduction
Examples	Email spam detection, house price prediction	Customer segmentation, grouping news articles
Evaluation Metrics	Accuracy, Precision, Recall, RMSE, etc.	Silhouette Score, Inertia, Davies–Bouldin Index, etc.
Algorithms	Linear Regression, Decision Trees, SVM, k-NN, Neural Networks	K-Means, Hierarchical Clustering, PCA, t-SNE, Autoencoders
Dependency on Labels	High (requires output labels for training)	None (no labeled output needed)

14. If Gradient Descent is diverging rather than converging, what is one possible technique to resolve this issue?

If **Gradient Descent is diverging** instead of converging, one common cause is a **learning rate that is too high**. A **large learning rate** can cause the algorithm to **overshoot the minimum** of the cost function, leading to increasing cost values and instability. Lowering the learning rate results in **smaller, more stable updates** to the parameters, which can help the algorithm converge properly.

15. What happens if we apply logistic regression to a dataset that is not linearly separable?

If we apply **logistic regression** to a dataset that is **not linearly separable**, the model may struggle to find a decision boundary that correctly classifies the data, leading to **poor performance**. High bias - Model underfits the data due to its limited capacity to model complex patterns.

16. How do you interpret the output probability of a logistic regression model?

The output of a **logistic regression** model is a **probability score between 0 and 1**, which represents the **likelihood** that a given input belongs to the **positive class (label = 1)**. If the model outputs **0.85**, it means there's an **85% probability** the input belongs to the positive class. If the model outputs **0.2**, it means there's a **20% probability**, so it's more likely to belong to the **negative class (label = 0)**.

17. Why is KNN called a lazy learning algorithm?

K-Nearest Neighbors (KNN) is called a **lazy learning algorithm** because it **does not learn a model during the training phase**. Instead, it **stores the entire training dataset** and only performs computation **at the time of prediction**.

18. How does the choice of 'K' affect the performance of the KNN algorithm?

The choice of 'K' in the **K-Nearest Neighbors (KNN)** algorithm critically affects its **performance, accuracy, and robustness**.

K Value	Effect
Small K	Model is sensitive to noise and outliers → High variance , may overfit.
Large K	Smoother decision boundary → Lower variance , but can underfit the data.
K too large	Includes too many irrelevant or distant neighbors → Poor class separation .

19. A company wants to classify whether a customer will purchase a product based on two features: Age Group and Income Level. For a given dataset, If a new customer is Middle-Aged with High Income, use the Naïve Bayes classifier to predict whether they will purchase the product.

Total samples = 9

- Purchase = Yes → 5
- Purchase = No → 4

For Purchase = Yes:

- $P(\text{Age} = \text{Middle-Aged} \mid \text{Yes}) = 2/5$
- $P(\text{Income} = \text{High} \mid \text{Yes}) = 1/5$
- $P(\text{Purchase} = \text{Yes}) = 5/9$

For Purchase = No:

- $P(\text{Age} = \text{Middle-Aged} \mid \text{No}) = 1/4$
- $P(\text{Income} = \text{High} \mid \text{No}) = 2/4$
- $P(\text{Purchase} = \text{No}) = 4/9$

For Purchase = **Yes**:

$$P(\text{Yes} \mid \text{Middle-Aged, High}) \propto P(\text{Yes}) \cdot P(\text{Middle-Aged} \mid \text{Yes}) \cdot P(\text{High} \mid \text{Yes}) = 5/9 \cdot 2/5 \cdot 1/5 = 2/45$$

For Purchase = **No**:

$$P(\text{No} \mid \text{Middle-Aged, High}) \propto P(\text{No}) \cdot P(\text{Middle-Aged} \mid \text{No}) \cdot P(\text{High} \mid \text{No}) = 4/9 \cdot 1/4 \cdot 2/4 = 1/18$$

$$P(\text{Yes}) \approx 0.044, P(\text{No}) \approx 0.056$$

Since **P(No) > P(Yes)**, the **prediction is: No, the customer will not purchase the product**.

Age_Group	Income_Level	Purchase (Yes/No)
Young	Low	No
Young	Medium	Yes
Young	High	No
Middle-Aged	Low	Yes
Middle-Aged	Medium	Yes
Middle-Aged	High	No
Senior	Low	No
Senior	Medium	Yes
Senior	High	Yes

20. A hospital is testing a new ML-based diagnostic tool on 100 patients. Of these, 50 have the disease, and 50 do not. The test correctly identifies 40 sick patients but misses 10. It also wrongly classifies 20 healthy patients as sick while correctly marking 30 as disease-free. Calculate the accuracy, precision, and recall of the ML-based diagnostic tool.

Given:

- **Total patients** = 100
- **Actually Sick** = 50
 - **Correctly Identified Sick (True Positives, TP)** = 40

- Missed Sick (False Negatives, FN) = 10
- Actually Healthy = 50
 - Wrongly Predicted Sick (False Positives, FP) = 20
 - Correctly Predicted Healthy (True Negatives, TN) = 30

	Predicted Sick	Predicted Healthy
Actually Sick	TP = 40	FN = 10
Actually Healthy	FP = 20	TN = 30

Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{40 + 30}{100} = \frac{70}{100} = \boxed{0.70 \text{ or } 70\%}$$

Precision (Positive Predictive Value):

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{40}{40 + 20} = \frac{40}{60} = \boxed{0.67 \text{ or } 66.7\%}$$

Recall (Sensitivity or True Positive Rate):

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{40}{40 + 10} = \frac{40}{50} = \boxed{0.80 \text{ or } 80\%}$$

21. A dataset consists of 16 instances, evenly distributed between two classes: Class A and Class B, with 8 instances each. After splitting the dataset based on attribute X, the data is divided into two subsets. The left node contains 6 instances, with 4 belonging to Class A and 2 to Class B, while the right node consists of 10 instances, with 4 from Class A and 6 from Class B. Calculate the information gain resulting from this split by using the entropy.

Calculate the entropy of the original dataset:

The formula for entropy $H(S)$ is:

$$H(S) = - \sum_{i=1}^k p_i \log_2 p_i$$

where p_i is the probability of class i in the dataset.

In this case, the dataset consists of 16 instances: 8 from Class A and 8 from Class B. Therefore, the probabilities for each class are:

$$p(\text{Class A}) = 8/16 = 0.5, \quad p(\text{Class B}) = 8/16 = 0.5$$

Thus, the entropy of the original dataset is:

$$H(\text{original}) = -(0.5 \log_2 0.5 + 0.5 \log_2 0.5) = -(0.5 \times (-1) + 0.5 \times (-1)) = 1$$

Calculate the entropy for the left and right subsets:

Left node:

- 6 instances: 4 from Class A, 2 from Class B.
- The probabilities for the left node are:

$$p(\text{Class A})=4/6=0.6667, \quad p(\text{Class B})=2/6=0.3333$$

The entropy for the left node is:

$$H(\text{left})=-(0.6667\log_2 0.6667+0.3333\log_2 0.3333)=-(0.6667\times(-0.58496)+0.3333\times(-1.58496))=0.9183$$

Right node:

- 10 instances: 4 from Class A, 6 from Class B.
- The probabilities for the right node are:

$$p(\text{Class A})=4/10=0.4, \quad p(\text{Class B})=6/10=0.6$$

The entropy for the right node is:

$$H(\text{right})=-(0.4\log_2 0.4+0.6\log_2 0.6)=-(0.4\times(-1.32193)+0.6\times(-0.73697))=0.9710$$

Calculate the weighted average of the entropies of the left and right nodes:

The weight for each node is based on the proportion of instances in each subset. The total number of instances is 16, so:

- Weight for the left node = $6/16=0.375$
- Weight for the right node = $10/16=0.625$

Thus, the weighted entropy is:

$$H(\text{split})=0.375\times 0.9183+0.625\times 0.9710=0.3446+0.6069=0.9515$$

Calculate the information gain:

Information gain is the difference between the entropy of the original dataset and the weighted entropy of the split:

$$IG=H(\text{original})-H(\text{split})=1-0.9515=0.0485$$

Thus, the information gain resulting from the split is **0.0485**.

22. What is Vanishing gradient?

The **vanishing gradient problem** occurs when gradients become very small during backpropagation, especially when training deep neural networks. This results in the model's weights not being updated properly, causing the learning process very slow.

23. What is Exploding gradient?

The **exploding gradient** occurs when the gradients during backpropagation become **too large**, results so many layers get large weight updates and the algorithm diverges.

24. What is gradient clipping?

Gradient clipping is a technique used to prevent the **exploding gradient problem** in deep neural networks by **limiting the size of the gradients** during backpropagation.

25. What is PCA (Principal Components Analysis) algorithm? Why is PCA used?

PCA (Principal Component Analysis) is a popular **dimensionality reduction** technique used in machine learning and data analysis to **reduce the number of features** (variables) in a dataset while **preserving as much variance** as possible.

Why PCA used?

- To **reduce complexity** in high-dimensional data.
- To **eliminate redundancy** caused by correlated features.
- To **visualize** high-dimensional data in 2D or 3D.
- To **speed up training** and reduce overfitting in ML models.

26. What is Factor Analysis (FA)? Why is FA used?

Factor Analysis (FA) is a **statistical technique** used to identify **underlying relationships** between observed variables. It assumes that **observable variables are influenced by a smaller number of unobservable variables**, and it tries to uncover those hidden factors.

Why FA used?

To **reduce data dimensionality** and **identify hidden patterns** or **latent constructs** that explain the correlations among a set of observed variables.